



Extensions

R2 implements some extensions on top of the basic S3 API. This page outlines these additional, available features. Some of the functionality described in this page requires setting a custom header. For examples on how to do so, refer to [Configure custom headers](#).

Extended metadata using Unicode

The [Workers R2 API](#) supports Unicode in keys and values natively without requiring any additional encoding or decoding for the `customMetadata` field. These fields map to the `x-amz-meta--` prefixed headers used within the R2 S3-compatible API endpoint.

HTTP header names and values may only contain ASCII characters, which is a small subset of the Unicode character library. To easily accommodate users, R2 adheres to [RFC 2047 ↗](#) and automatically decodes all `x-amz-meta-*` header values before storage. On retrieval, any metadata values with unicode are RFC 2047-encoded before rendering the response. The length limit for metadata values is applied to the decoded Unicode value.

Metadata variance

Be mindful when using both Workers and S3 API endpoints to access the same data. If the R2 metadata keys contain Unicode, they are stripped when accessed through the S3 API and the `x-amz-missing-meta` header is set to the number of keys that were omitted.

These headers map to the `httpMetadata` field in the [R2 bindings](#):

HTTP

Property Name

Header

Content-
Encoding

`httpMetadata.contentEncoding`

Content-
Type

`httpMetadata.contentType`

Content-
Language

`httpMetadata.contentLanguage`

Content-
Disposition

`httpMetadata.contentDisposition`

Cache-
Control

`httpMetadata.cacheControl`

Expires

`httpMetadata.expires`

If using Unicode in object key names, refer to [Unicode Interoperability](#).

Auto-creating buckets on upload

If you are creating buckets on demand, you might initiate an upload with the assumption that a target bucket exists. In this situation, if you received a `NoSuchBucket` error, you would probably issue a `CreateBucket` operation. However, following this approach can cause issues: if the body has already been partially consumed, the upload will need to be aborted. A common solution to this issue, followed by other object storage providers, is to use the [HTTP 100](#) response to detect whether the body should be sent, or if the bucket must be created before retrying the upload. However, Cloudflare does not support the HTTP 100 response. Even if the HTTP 100 response was supported, you would still have additional latency due to the round trips involved.

To support sending an upload with a streaming body to a bucket that may not exist yet, upload operations such as `PutObject` or `CreateMultipartUpload` allow

you to specify a header that will ensure the `NoSuchBucket` error is not returned. If the bucket does not exist at the time of upload, it is implicitly instantiated with the following `CreateBucket` request:

```
PUT / HTTP/1.1
Host: bucket.account.r2.cloudflarestorage.com
<CreateBucketConfiguration
  xmlns="http://s3.amazonaws.com/doc/2006-03-01/">

  <LocationConstraint>auto</LocationConstraint>
</CreateBucketConfiguration>
```

This is only useful if you are creating buckets on demand because you do not know the name of the bucket or the preferred access location ahead of time. For example, you have one bucket per one of your customers and the bucket is created on first upload to the bucket and not during account registration. In these cases, the [ListBuckets extension](#), which

supports accounts with more than 1,000 buckets, may also be useful.

PutObject and CreateMultipartUpload

cf-create-bucket-if-missing

Add a `cf-create-bucket-if-missing` header with the value `true` to implicitly create the bucket if it does not exist yet. Refer to [Auto-creating buckets on upload](#) for a more detailed explanation of when to add this header.

PutObject

Conditional operations in PutObject

PutObject supports [conditional uploads](#) via the [If-Match](#), [If-None-Match](#), [If-Modified-Since](#), and

[If-Unmodified-Since ↗](#) headers. These headers will cause the PutObject operation to be rejected with 412 PreconditionFailed error codes when the preceding state of the object that is being written to does not match the specified conditions.

CopyObject

MERGE metadata directive

The `x-amz-metadata-directive` allows a MERGE value, in addition to the standard COPY and REPLACE options. When used, MERGE is a combination of COPY and REPLACE, which will COPY any metadata keys from the source object and REPLACE those that are specified in the request with the new value. You cannot use MERGE to remove existing metadata keys from the source — use REPLACE instead.

ListBuckets

ListBuckets supports all the same search parameters as ListObjectsV2 in R2 because some customers may have more than 1,000 buckets. Because tooling, like existing S3 libraries, may not expose a way to set these search parameters, these values may also be sent in via headers. Values in headers take precedence over the search parameters.

Search parameter	HTTP Header	Meaning
prefix	cf-prefix	Show buckets with this prefix only.
start-after	cf-start-after	Show buckets whose name appears lexicographically in the account

Search parameter	HTTP Header	Meaning
continuation-token	cf-continuation-token	Resume listing from a previously returned continuation token.
max-keys	cf-max-keys	Return this maximum number of buckets. Default and max is 10

The XML response contains a `NextContinuationToken` and `IsTruncated` elements as appropriate. Since these may not be accessible from existing S3 APIs, these are also available in response headers:

XML Response Element	HTTP Response Header	Meaning
IsTruncated	cf-is-truncated	This header indicates whether the list of objects returned by the GET request is truncated. If the value is true, the list of objects is truncated and the next continuation token must be used to retrieve the next page of objects.
NextContinuationToken	cf-next-continuation-token	This header contains the next continuation token to use in the next GET request to retrieve the next page of objects. The token is a base64-encoded string that represents the position of the next object in the list.

XML Response Element	HTTP Response Header	Meaning
StartAfter		This start value was in or requ
KeyCount		The of b retu
ContinuationToken		The cont toke was supp the r

XML Response Element	HTTP Response Header	Meaning
MaxKeys		The maximum number of keys returned in the response. If not specified, the response will return all keys.

Conditional operations in CopyObject for the destination object

Note

This feature is currently in beta. If you have feedback, reach out to us on the [Cloudflare Developer Discord ↗](#) in the #r2-storage channel or open a thread on the [Community Forum ↗](#).

CopyObject already supports conditions that relate to the source object through the `x-amz-copy-source-if-...` headers as part of our compliance with the S3 API. In addition to this, R2 supports an R2 specific set of headers that allow the CopyObject operation to be conditional on the target object:

- `cf-copy-destination-if-match`
- `cf-copy-destination-if-none-match`
- `cf-copy-destination-if-modified-since`
- `cf-copy-destination-if-unmodified-since`

These headers work akin to the similarly named conditional headers supported on PutObject. When

the preceding state of the destination object to does not match the specified conditions the CopyObject operation will be rejected with a 412 PreconditionFailed error code.

Non-atomicity relative to x-amz-copy-source-if

The x-amz-copy-source-if-... headers are guaranteed to be checked when the source object for the copy operation is selected, and the cf-copy-destination-if-... headers are guaranteed to be checked when the object is committed to the bucket state. However, the time at which the source object is selected for copying, and the point in time when the destination object is committed to the bucket state are not necessarily the same. This means that the cf-copy-destination-if-... headers are not atomic in relation to the x-amz-copy-source-if... headers.

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